
UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 8: The Discrete Fourier Transform (DFT)

Lecturer Prof. Justin Haldar

Collected by Gewenxin Yu

Date September 17, 2026

1 The Discrete Fourier Transform (DFT)

Def.

The DFT is a mapping of a length- N sequence $x[n]$, $0 \leq n \leq N - 1$, into a length- N Fourier representation $X[k]$, $k = 0, \dots, N - 1$, according to

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}.$$

It is a **sampled version of the DTFT** for a finite-length sequence:

$$X[k] = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}}.$$

$\Rightarrow$ discrete in both time and frequency .

Inverse DFT.

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{+j \frac{2\pi kn}{N}}, \quad n = 0, \dots, N - 1.$$

2 1.1 Twiddle-Factor Notation

With $W_N = e^{-j\frac{2\pi}{N}}$,

$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1, \\ x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq n \leq N-1. \end{aligned}$$

Verification of the inverse formula.

$$\begin{aligned} X[\ell] &= \sum_{n=0}^{N-1} x[n] W_N^{\ell n} = \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \right] W_N^{\ell n} \\ &= \sum_{k=0}^{N-1} X[k] \underbrace{\left[\frac{1}{N} \sum_{n=0}^{N-1} W_N^{-(k-\ell)n} \right]}_{(*)}. \end{aligned}$$

Summing the geometric series,

$$\begin{aligned} (*) &= \frac{1}{N} \frac{1 - W_N^{-(k-\ell)N}}{1 - W_N^{-(k-\ell)}} = \frac{1}{N} \frac{1 - e^{j2\pi(k-\ell)}}{1 - e^{j\frac{2\pi(k-\ell)}{N}}} \\ &= \frac{1}{N} \frac{e^{j\pi(k-\ell)} (e^{-j\pi(k-\ell)} - e^{j\pi(k-\ell)})}{e^{j\frac{\pi(k-\ell)}{N}} (e^{-j\frac{\pi(k-\ell)}{N}} - e^{j\frac{\pi(k-\ell)}{N}})} \\ &= \frac{1}{N} \frac{\sin(\pi[k-\ell]) e^{j\pi(k-\ell)}}{\sin(\frac{\pi}{N}[k-\ell]) e^{j\frac{\pi(k-\ell)}{N}}} \end{aligned}$$

3 Hence

$$(*) = \begin{cases} 1, & k - \ell = rN, \text{ } r \text{ integer,} \\ 0, & \text{else} \end{cases} = \delta[\langle k - \ell \rangle_N].$$

Remark*(addendum to orig lecture note).*

The geometric-series formula is only valid when $W_N^{-(k-\ell)} \neq 1$, i.e., $k-\ell \neq rN$; in that case $\sin(\pi[k-\ell]) = 0$ while $\sin(\frac{\pi}{N}[k-\ell]) \neq 0$, so $(*) = 0$. When $k-\ell = rN$, every term of the sum equals $W_N^{-rNn} = 1$, so $(*) = \frac{1}{N} \cdot N = 1$ directly (this is also the $0/0$ limit of the Dirichlet-kernel expression).

Modulo Operation $\langle n \rangle_N$.

$k = \langle n \rangle_N$ if $k \in \{0, \dots, N-1\}$ and there exists an integer ℓ such that $n = k + \ell N$.

Example ($N = 3$).

n	-2	-1	0	1	2	3	4	5	6
$\langle n \rangle_3$	1	2	0	1	2	0	1	2	0

Therefore

$$X[\ell] = \sum_{k=0}^{N-1} X[k] \delta[\langle k - \ell \rangle_N] = X[\ell] \quad \text{for } \ell \in \{0, \dots, N-1\}. \quad \blacksquare$$

4

Example.

$$g[n] = \cos\left(\frac{2\pi rn}{N}\right), \quad 0 \leq n \leq N-1, \text{ with } r \text{ an integer, } 0 \leq r \leq N-1.$$

$$g[n] = \frac{1}{2} (W_N^{-rn} + W_N^{rn})$$

$$\begin{aligned} \Rightarrow G[k] &= \frac{1}{2} \left(\sum_{n=0}^{N-1} W_N^{-(r-k)n} + \sum_{n=0}^{N-1} W_N^{-(r+k)n} \right) \\ &= \frac{1}{2} \left(N \delta[\langle r-k \rangle_N] + N \delta[\langle -r-k \rangle_N] \right). \end{aligned}$$

For $0 \leq k \leq N-1$:

$$G[k] = \begin{cases} \frac{N}{2}, & k = r, \quad k = N-r, \\ 0, & \text{else,} \end{cases} \quad (\text{unless } r = 0 \text{ or } r = N/2)$$

using $\langle -r-k \rangle_N = \langle N-r-k \rangle_N$. When $r = 0$ or $r = N/2$, the two impulses coincide and $G[r] = N$.

5 2 DFT Properties

1

2.1 Periodicity

Property.

$$\begin{aligned} X[\ell N + k] &= X[k], & k = 0, \dots, N-1, \ell \text{ an integer,} \\ x[mN + n] &= x[n], & n = 0, \dots, N-1, m \text{ an integer,} \end{aligned}$$

when the DFT / IDFT formulas are evaluated outside $0, \dots, N-1$.

¹cf. Mitra Tables 5.1, 5.2, and 5.3 (reproduced in the appendix of the full build).

Because

$$W_N^N = e^{-j\frac{2\pi}{N}N} = e^{-j2\pi} = 1 \quad \Rightarrow \quad W_N^{\ell N+k} = (1)^\ell W_N^k = W_N^k.$$

2.2 Circular Shift

Consider $x[n]$, $0 \leq n \leq N-1$. The ordinary shift $x_1[n] = x[n-n_0]$ is **no longer defined** over $0 \leq n \leq N-1$.

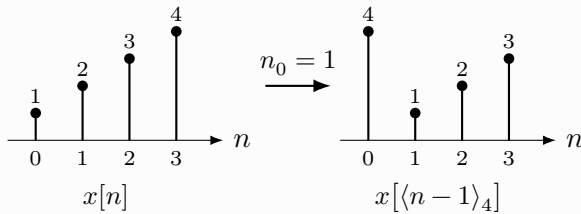
$$\Rightarrow \text{circular shifts:} \quad x_1[n] = x[\langle n - n_0 \rangle_N].$$

6

Example (Right Circular Shift).

Let $0 < n_0 \leq N-1$:

$$x_1[n] = x[\langle n - n_0 \rangle_N] = \begin{cases} x[n - n_0], & n_0 \leq n \leq N-1, \\ x[N + n - n_0], & 0 \leq n \leq n_0 - 1. \end{cases}$$



$$x[n] = \{1, 2, 3, 4\}, \quad x[\langle n-1 \rangle_4] = \{4, 1, 2, 3\},$$

$$x[\langle n-2 \rangle_4] = \{3, 4, 1, 2\} = x[\langle n+2 \rangle_4].$$

Property (Circular Time-Shifting).

Let $g[n] \leftrightarrow G[k]$ and $\tilde{g}[n] = g[\langle n - n_0 \rangle_N]$. Then

$$\tilde{G}[k] = W_N^{kn_0} G[k] = e^{-j\frac{2\pi kn_0}{N}} G[k].$$

7**Proof.**

$$\tilde{G}[k] = \sum_{n=0}^{N-1} g[\langle n - n_0 \rangle_N] W_N^{kn} = \sum_{n=0}^{n_0-1} g[n + N - n_0] W_N^{kn} + \sum_{n=n_0}^{N-1} g[n - n_0] W_N^{kn}.$$

Let $m = n + N - n_0$ ($n = m - N + n_0$) in the first sum and $p = n - n_0$ ($n = p + n_0$) in the second:

$$\begin{aligned} \tilde{G}[k] &= \sum_{m=N-n_0}^{N-1} g[m] W_N^{k(m-N+n_0)} + \sum_{p=0}^{N-1-n_0} g[p] W_N^{k(p+n_0)} \\ &= \sum_{n=0}^{N-1} g[n] W_N^{k(n+n_0)} \quad (W_N^{-kN} = 1) \\ &= W_N^{kn_0} \left[\sum_{n=0}^{N-1} g[n] W_N^{kn} \right] = W_N^{kn_0} G[k] = e^{-j\frac{2\pi kn_0}{N}} G[k]. \quad \blacksquare \end{aligned}$$

8**2.3 Convolution**

Linear convolution and the DTFT:

$$x[n] \otimes h[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) H(e^{j\omega}).$$

If $x[n]$ has length N_1 and $h[n]$ has length N_2 , then $y[n] = x[n] \otimes h[n]$ has length $N_1 + N_2 - 1$: it is possibly nonzero for $0 \leq n \leq N_1 + N_2 - 2$ and is zero otherwise.

Circular Convolution (two length- N sequences).

$$y_c[n] = \sum_{m=0}^{N-1} x[m] h[\langle n - m \rangle_N], \quad 0 \leq n \leq N - 1,$$

$$y_c[n] = x[n] \circledcirc h[n] = h[n] \circledcirc x[n].$$

Convolution Theorem.

If $y_c[n] = x[n] \circledcirc h[n]$, then $Y_c[k] = X[k] H[k]$.

9**Proof.**

$$\begin{aligned} Y_c[k] &= \sum_{n=0}^{N-1} \left[\sum_{m=0}^{N-1} x[\langle n - m \rangle_N] h[m] \right] W_N^{kn} = \sum_{m=0}^{N-1} h[m] \left[\sum_{n=0}^{N-1} x[\langle n - m \rangle_N] W_N^{kn} \right] \\ &= \left[\sum_{m=0}^{N-1} h[m] W_N^{km} \right] X[k] \quad (\text{circular time-shift}) \\ &= H[k] X[k]. \quad \checkmark \end{aligned}$$

2.4 Linear Convolution via Circular Convolution

If $x[n], 0 \leq n \leq N_1 - 1$, and $h[n], 0 \leq n \leq N_2 - 1$, then

$$y[n] = x[n] \circledast h[n], \quad 0 \leq n \leq N_1 + N_2 - 2.$$

Property.

If

$$\tilde{x}[n] = \begin{cases} x[n], & 0 \leq n \leq N_1 - 1, \\ 0, & N_1 \leq n < N, \end{cases} \quad \text{where } N \geq N_1 + N_2 - 1,$$

and similarly define a “zero-padded” length- N $\tilde{h}[n]$, then

$$\tilde{x}[n] \circledcirc \tilde{h}[n] = x[n] \circledast h[n] \quad \text{for } 0 \leq n \leq N - 1.$$

Consequence: we can compute $\circledast$ using $\circledcirc N!$.

1. Zero-pad both sequences to the same length $N \geq N_1 + N_2 - 1$.
2. Take length- N DFTs of both.
3. Multiply the DFTs together.
4. Take the inverse DFT of the result.

Remark

(addendum to orig lecture note).

Why $N \geq N_1 + N_2 - 1$: circular convolution equals the linear convolution *aliased* with period N , $\tilde{x}[n] \circledcirc \tilde{h}[n] = \sum_r y[n + rN]$ for $0 \leq n \leq N - 1$. Since $y[n]$ occupies $N_1 + N_2 - 1$ samples, no shifted copies overlap exactly when $N \geq N_1 + N_2 - 1$. With a fast algorithm (FFT) for the DFT, this is how long convolutions are computed in practice (cf. Mitra Ch. 11).

2.5 Parseval's Theorem

Property.

$$\sum_{n=0}^{N-1} x^*[n] y[n] = \frac{1}{N} \sum_{k=0}^{N-1} X^*[k] Y[k] \quad \Rightarrow \quad \sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2.$$

2.6 Symmetry

Property.

If $x[n]$ is real,

$$X[k] = X^*[N - k] = X^*[\langle -k \rangle_N], \quad 0 \leq k \leq N - 1.$$

(For $k = 0$, read $X^*[N]$ as $X^*[0]$ by periodicity; the modulo form handles this automatically.)

Readings

- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Ch. 5.4, 5.5, 5.7, 11.3–11.5
- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 6, 7